Unmanned Ground Vehicle with Real-Time Object Detection Audio Communication and Controlled Weapon Integration

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***Abstract* -- Unmanned Ground Vehicles (UGVs) are being developed to do surveillance and perform hazardous activities with the minimum amount of risk to humans. This paper describes the development of a UGV with real-time object detection, audio communication, and controlling a weapon. An edge-AI-based method allows for the visual processing done onboard without relying on a cloud, so low latency and dependable operation can be ensured. Onboard cameras perform with a deep learning model to detect people and obstacles in real-time; ultrasonic sensors further support navigation and collision avoidance. Two-way audio communication allows for both the operator and environment to communicate with one another. The weapon system is integrated and used only after being authorized by the operator. GPS tracking makes it possible to continuously track the location of the UGV, which creates opportunities to use it in numerous applications such as surveillance, security, and rescue operations.**

Keywords – Unmanned Ground Vehicle, Object Detection, YOLOv8, Computer Vision, Audio Communication, Autonomous Robotics.

I. INTRODUCTION

Unmanned Ground Vehicles are robotic platforms designed for operation on the ground without human presence. A wide distribution of UGVs is presently attained in defense, security, industrial inspection, disaster management, and surveillance applications. Traditional ground robots are highly dependent on manual control with limited sensing, which seriously constrains their operational effectiveness in dynamic environments.

Recent advancements made possible through embedded systems, computer vision, and deep learning techniques have made UGV technology able to perceive and understand its environment. The object detection capability of robots helps detect humans, vehicles, and obstacles, while wireless communication technology makes it possible to monitor and control the robot remotely. But these days, most remote-control robots only use visual interaction, which does not support audio interaction, an essential task when conducting rescue or security missions.

A proposed system in this regard is the UGV with real-time object detection and two-way audio communication. Here, the system incorporates the use of a camera-based deep learning model for object detection along with an audio module for communication between the operator and the environment. Hence, the system promotes autonomy in the UGV while also enabling human interaction functionality.

II. METHODOLGY

A new technology will incorporate an unmanned ground vehicle (UGV) that incorporates edge-based artificial intelligence technologies for the ability to see things in real-time and also allows for two-way audio interactions with the operator. The UGV uses microcontroller technology to control its movement with DC motors and a motor driver module, allowing for differential drive navigation. In addition to video capture, the UGV has a monocular camera mounted to capture video continuously, and the video is processed locally by the device running edge AI, so that the UGV does not need to rely on Internet access or wait for processing; both of these features allow the UGV to have a real-time view of the environment. A lightweight deep learning object detection model, which was trained for deployment on an edge device, is used to detect obstacles, people, and target objects as they pass in front of the UGV. The UGV uses this information to aid in navigation and situation assessment. The audio component of the device consists of a microphone and speaker module connected to the processing unit with wireless communication links, to allow two-way voice communication between the operator and the UGV. The UGV's edge devices will communicate to one another via Wi-Fi links to transmit detection results, and receive command instructions from a remote control interface. The audio signals, sensor data, and visual perception results are synchronized to improve the reliability of operation in a variety of dynamic situations. Combined, the above methods allow the UGV to perform autonomous perception, interactive communication, and

remote monitoring in an efficient manner at the edge of the Internet.

III. EXISTING METHOD

This study provides an explanation of the real-time implementation of the proposed framework in the form of an executable code as per the developed method in identifying ground objects from the images taken from the UAVs. Consequently, the study provides an explanation of the detection of the excavators in the pipeline environment. This method relies on the implementation of the CNN in the detection of the objects in the images received from the UAVs. Despite the efficiency of the proposed method in the detection of objects in the images from the aerial view of the environment, this method is not suitable in the implementation of the autonomous ground vehicles due to its requirement for the UAVs in the detection of objects in the environment. [1]

A paper proposes a cloud-based system that utilizes a cloud-based system that focuses on detecting objects in real-time by sending video information from UAVs to the cloud, where information will be processed considering deep learning approaches. Considering that the proposed system utilizes a scalable computing ability, together with its enhanced precision, a negative aspect revolves around the fact that such a system will experience connection limitations, where communication will be required at all times.

Such a system may not be applicable in an EGVC competition setting, considering that it heavily relies on cloud-based infrastructure.[2]

This study presents a multimodal approach using a multimodal transformer model in the detection of UAV video using a combination of radar, audio, and video signals. This enhances the robustness of the detection through the use of sensor fusion techniques. This approach, although robust, is quite powerful in terms of computation. It is, however, focused on the recognition of objects from the air. It, therefore, limits the use of UGV.[7]

This paper presents an audio-vision-based emergency vehicle detection technique that incorporates both sound source localization and visual object recognition to effectively address the challenges involved in detecting objects in complex environments such as cities and roads. This technique will use audio signals for detecting distinctive patterns of vehicle sirens to identify their positions and then use video for object detection to validate images pertaining to emergency vehicle identification. This technique will be useful for detecting objects with higher degrees of precision than merely relying on either of these two sensors by itself to tackle the complexities in dealing with heavy traffic and high noise situations that may prevail during emergency scenarios.

Nevertheless, this technique appears to be suitable for traffic identification as well as traffic management based on fixed camera positions to sense objects moving with specially fixed speeds, as no provisions exist to tackle issues pertinent to the navigation of robots that may be employed in ground vehicle surveillance systems as part of an unmanned ground vehicle setup for surveillance purposes.[10]

This paper presents a summary of deep learning-based object detection methodologies that have been implemented on unmanned aerial vehicles with an embedded computing platform. The proposed methodology aims at performing onboard processing of visual data through embedded hardware, thereby enabling real-time inference and minimizing the use of cloud-based infrastructure to reduce latency in communication. The deployment of deep learning models on UAVs makes the system highly responsive with operational reliability in cases of limited network connectivity. Even with this distinct advantage, the proposed framework is built for aerial platforms with an assumption of flight-based sensing, wide-area visual coverage, and motion dynamics pertaining to UAVs. The systems do not deal with constrained field-of-view issues and other challenges of unmanned ground vehicles, like interaction with obstacles on the ground, variability in terrain, and real-time navigation control. Therefore, while the strategy of embedded processing is feasible and operative for UAV applications, their extension to UGVs is not possible without significant architectural and algorithmic changes. [12]

This paper focuses on the application of the UAS technique in improving the overall safety conditions on construction sites though monitoring and inspection activities carried out on the site. This technique involves the monitoring or inspection of the construction site or area with the help of drones or UAVs. Such monitoring or inspection enables the identification or detection of several hazardous conditions or parameters on the site or area. Although the proposed technique appears viable with regard to providing high-level visual inspection and analysis of the overall safety on the construction site or area, the approach relies on the acquisition of data only with the help of drones or flying devices. Thus, the proposed technique lacks the ability or the scope to integrate or acquire data with the help of the ground or surface level. Hence, the proposed approach cannot be used or implemented on the construction site or area with regard to the task or application of autonomous safety monitoring. [18]

IV.RESEARCH GAP

By analyzing the existing methods for unmanned systems, it has been determined that current research is fragmented into various areas and does not provide a single, cohesive automated solution for all areas of unmanned systems development. Several publications have been produced that deal with UAV (unmanned aerial vehicle) assisted surveillance applications, as well as IoT (internet of things) based image data collection in which data is transmitted securely to a centralised location for data processing. While these UAV methods and systems are effective in monitoring a given area from above, they use a lot of cloud infrastructure for processing and transferring of data which creates a latency and reliability issue in communication restricted and hostile environments. Furthermore, these systems are primarily air-based and do not solve any of the problems pertaining to unmanned ground vehicles such as obstacle avoidance, close-range interactions, and autonomous navigation required for unmanned ground vehicles.

IoT based UGV control systems provide remote access to navigation and monitoring of UGVs, however, they are largely dependent on centralised decision-making and operator intervention, therefore not providing the capability to make decisions quickly and effectively in a dynamic and unpredictable environment. Due to the lack of onboard intelligence and real-time perception, IoT based UGV control systems are not able to react quickly in a given circumstance and are made irrelevant in situations requiring speedy local decision-making. Similarly, AI-based vision enhancement systems, which demonstrate how developing embedded image processors and processing will make drones capable of doing many tasks without requiring any type of operator intervention or onboard computer systems for processing, are limited to fixed/static and weapon-mounted platforms, therefore they are not able to accommodate a UGV's necessity to be mobile, navigate, and interact with the environment in which it operates.

Several studies have examined the concept of intelligent automation and edge AI and described the advantages of Deep - based AI processing models developed with machine learning techniques for the development of automated unmanned systems.

V.PROPOSED METHODOLOGY

A. Edge-AI Based Unmanned Ground Vehicle Architecture

The proposed system is an Edge-AI enabled Unmanned Ground Vehicle (UGV) intended to be used in real-time surveillance and monitoring in both daylight as well as low-light conditions. The system consists of an embedded controller that is capable of processing visual, audio, as well as location information without depending on cloud services. The proposed system is highly optimized in terms of latency due to its edge computing capabilities, ensuring it works well in either limited or hostile communication environments. The UGV is also self-contained in nature, consisting of sensing, computing, communication, and control components that are capable of traversing challenging paths while monitoring their surroundings consistently.

The system's architectural structure consists of a rover base chassis that is propelled by DC motors, an embedded processing mechanism, an image processing module that enables real-time processing of objects in focus, and other embedded sensors that facilitate monitoring of audio as well as tracking..

B. Real-Time Object Detection Using ESP32-CAMg

Real-time object detection and analysis are performed through the use of the ESP32-CAM module. This module acts as the visual sensor for the UGV. The sensor continuously captures video from its surroundings and transmits it to the processing unit for analysis. An object detection system based on supervised machine learning algorithms is implemented at the edge for objects such as humans and obstacles.

For facilitating night vision surveillance capabilities, infrared LEDs are integrated with the ESP32-CAM module, which provides night vision capabilities. The images captured are processed to identify significant features, to which bounding boxes are applied. This allows embedded object detection without relying on cloud computing, which is crucial in military scenarios due to its sensitivity to time.

C. Autonomous Navigation and Audio Communication Framework

The UGV is intended to run autonomously, working with sensor-based navigation and intelligent decision-making. The ultrasonic sensors used within the robot constantly scan the area to detect any object that might help in the process of avoiding a collision. Depending on the sensor signals, the robot’s controller alters the direction of the motor to achieve a smooth path.

Besides the visual detection capabilities, the system also has real-time audio communication. This enhances the situation awareness. The audio communication module consists of a microphone that records the sounds from the surroundings. These sounds give the operator the ability to listen to the audio at a distant location. The audio communication framework also allows the operator to give commands through voice interaction. The operator sends commands to the

UGV from the control station.

The autonomous navigation system is intended to dynamically adjust to the environment by integrating the sensing data with the perception results in real time. The UGV will ensure the highest safety margin during traversal by modulating the motion based on the proximity to the obstacles and the nature of the objects in the environment. The navigation commands are processed at the edge to ensure immediate execution without any delay from external computation. On the other hand, the audio communication subsystem is concurrent with the navigation operations, allowing the continuous capturing of environmental sounds and user voice interaction during traversal.

D. Weapon Integration and Controlled Actuation Module

Alongside surveillance and navigation capabilities, the proposed Unmanned Ground Vehicle also incorporates an optional weapon integration module to further increase the utility of the system for defense and security applications. A lightweight gun assembly is mounted onto the UGV chassis and integrated with the on-board controller via a controlled actuation device. The firing mechanism is activated via an electrically actuated relay or servo-based trigger mechanism, allowing for accurate and safe operation. By default, the weapon is not autonomous and can only fire under the specific control commands issued by the operator via the communications interface. The Edge-AI based object detection module provides visual feedback to the operator to help identify potential targets for engagement; gun actuation will only occur after explicit command authorization is given. Thus, weapon integration remains as a supportive defensive element, maintaining overall system safety, navigation stability, and real-time perception. The modular design of the weapon system allows it to be enabled or disabled based upon the mission requirements, thereby allowing for user flexibility in operating UGVs for both civilian surveillance and defense-deployed operations.

E. Proposed UGV System Block Diagram

The following Fig. 1 displays the block diagram of the proposed UGV system.

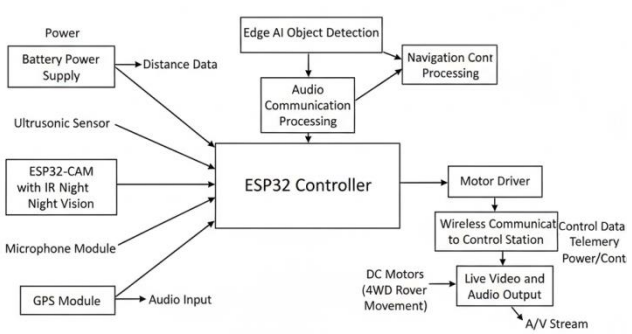

Fig. 3.1. Proposed System UGV Block diagram

The following Fig. 3.2 illustrates the integration of the Edge-AI based real-time object detection module with the ESP32 controller in the Unmanned Ground Vehicle system.

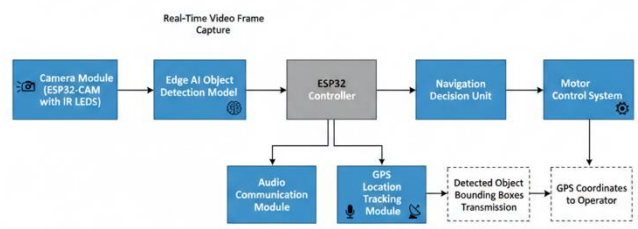

Fig. 3.2. Integration of Edge-AI based Real-Time Object Detection with ESP32 Controller

In this regard, Figure 5 provides a system block diagram that indicates how hardware and software components are linked to work effectively within a UGV. The embedded controller serves as a CPU that interacts with each respective unit such that it receives video and object data from the ESP32-CAM module. Additionally, data from ultrasonic sensors and audio signals from the audio module are received.

The GPS receiver is always updating the geographical location information for the UGV and the targets identified. The motor drivers are controlled by the control signals from the controller in order to control the movement of the rover. All the processed data from the image stream, detection outputs, auditory signals, and the GPS locations are sent to the operator interface for analysis. The block diagram explains the control logic and data flow in the complete system.

VI.SYSTEM IMPLEMENTATION

A. *Embedded Hardware and Software Environment Setup*

The hardware components involve a 4WD rover, which consists of DC motors for movement. There is also an embedded controller. The ESP32-CAM is mounted on the rover to ensure a continuous flow of images. Sensors are also included to facilitate detection. A microphone and a GPS module are integrated to allow sound tracking.

The software environment is setup using an embedded development framework, which is compatible

with ESP32 microcontrollers. All the required libraries for camera interaction, motor operations, sensor communication, and wireless data communication have also been initialized. The firmware designed for the system supports the execution of multiple tasks related to image capture, object detection, navigation, audio, and GPS data in real time.

B. *Real-Time Object Detection Model Initialization*

The object detection engine is initialized when the system is initiated by loading pre-trained parameters, which are optimized for embedded environment performance. The pre-trained parameters are created through training the engine on a set of images of human, vehicle, and environmental obstacle classes, which are captured in varying lighting conditions. Before executing a detection, image frames are pre-processed.

After being initialized, it keeps executing the inference task of analyzing incoming video frames for real-time detection and tracking of the objects. The detected objects are marked by the addition of boundary boxes around them. The required information regarding the detection of the objects is also transferred to the navigation and alert system.

To make the performance on the resource-limited embedded system robust, the parameters for the object detection model are optimized in the initialization process for lightweight network parameters and efficient inference configuration. Memory allocation, processing, and other parameters are set to achieve a trade-off between the accuracy of the object detection process and the real-time capability of the system. Also, the initialization process entails the validation of input from the camera, the frame rate regulation, as well as the regulation of the confidence threshold in the object detection system.

Fig.4.1 Implementation of the Proposed system

The above Fig. 4.1 displays the implementation of the proposed system.

C. *Edge-AI Integration with Audio and GPS Communication*

The integration of edge-AI technology enables the ugv to seamlessly combine visual intelligence with audio and location intelligence. Objects identified are linked to actual location information through GPS, and audio information collected from the environment can be streamed together with video and detection information.

Processing all the information at the edge will ensure no delays in communication. The end-to-end solution provides a platform where the UGV is able to make decisions concerning navigation and alerts simultaneously. Additionally, it relays vital information to the control station. The solution combines AI perception, sound communications, and GPS tracking to improve the functionality of the surveillance solution.

VII.CONCLUSION

This project described the design & development of the Unmanned Ground Vehicle that uses Edge AI to perform real-time object detection, self-navigation, audio communication, GPS location tracking, & more. The proposed design minimizes direct human interaction in dangerous settings & offers much-improved situational awareness capabilities. The Edge AI ensures low latency & reliance upon the cloud.

Future upgrades could be in the form of path planning algorithms, coordinating UGVs, range communication systems, and upgrading the deep learning models to achieve better detection precision. The proposed system can be extended in the field of disaster management and search and rescue.

The proposed UGV design is scalable. It can be scaled to support different applications. In fact, the design will require only minor adjustments to be used for other missions such as search and rescue, perimeter protection of industrial estates, and disaster response. Future studies could involve applying the latest edge-optimized deep learning algorithms, range communication technologies, or collaborative UGV architectures to promote increased coverage and protection. Inclusion of autonomous route planning algorithms or efficient power control strategies may also be considered.

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